



OPEN Double 3 MJ dense plasma focus for thermonuclear drive inertial confinement fusion

S. M. Sadat Kiai¹✉, Shirin Adlparvar², Hossein Sadeghi^{1,4} & Samaneh Fazelpour³

Nuclear fusion remains one of the most promising solutions for clean and sustainable energy production. However, significant challenges—including energy losses, plasma instabilities, and high operational costs—continue to hinder its practical realization. While magnetic confinement fusion (MCF) and inertial confinement fusion (ICF) have been the dominant approaches, dense plasma focus (DPF) devices present a compact and high-performance alternative. This study introduces a novel double-DPF system, employing two coaxial DPF devices to compress and accelerate deuterium-tritium (DT) fuel pellets, leading to enhanced energy transfer and ignition conditions. By integrating high-temperature superconducting (HTS) magnetic field lenses, the proposed system significantly improves plasma confinement, suppresses turbulence, and enhances fusion efficiency. Key physical processes—including pinch dynamics, confinement time enhancement, preheating mechanisms, and neutron yield estimations—are rigorously analyzed through magnetohydrodynamic (MHD) models and numerical simulations. Theoretical results suggest that HTS-assisted double-DPF operation triples the fusion power output compared to conventional single-DPF configurations. Furthermore, optimized energy coupling between the plasma and the DT target enhances the probability of achieving ignition conditions. This work provides a systematic theoretical framework that lays the foundation for future laboratory validation. While further experimental and engineering studies are necessary, the double-DPF concept represents a scalable and efficient pathway toward controlled thermonuclear fusion. By addressing confinement and energy transfer limitations, this study contributes to the ongoing pursuit of practical fusion-based energy generation.

Keywords Pulsed power, Dense plasma focus, Thermonuclear reaction mechanism, HTS magnetic field lenses, Target pellet, Implosion, Fusion yield

Nuclear fusion, the process by which two light atomic nuclei combine to form a heavier nucleus, has the potential to provide a nearly limitless, clean energy source. This process powers stars, including our sun, and is driven by the release of significant amounts of energy when the nuclear binding energy of the resulting heavier nucleus exceeds that of the initial nuclei. Achieving controlled nuclear fusion on earth, however, poses formidable scientific and engineering challenges. For fusion to occur, the coulomb barrier—electrostatic repulsion between positively charged nuclei—must be overcome. This requires extremely high temperatures (on the order of millions of degrees) and pressures to bring nuclei close enough for the strong nuclear force to dominate and initiate fusion. Researchers have developed several approaches to replicate these conditions, with the two leading methods being inertial confinement fusion (ICF)^{1–5}, and magnetic confinement fusion (MCF)^{6–10}, each of which has made significant strides toward practical energy generation. Inertial confinement fusion (ICF), relies on compressing and heating a small fuel pellet, typically composed of deuterium and tritium (DT), to achieve the conditions necessary for fusion. This is achieved using high-energy lasers, particle beams (light or heavy ions), or other devices. Two primary techniques exist within ICF, direct drive and indirect drive. In direct drive, lasers or particle beams directly strike the pellet's surface, compressing it symmetrically to achieve fusion conditions. In indirect drive, energy from lasers or particle beams heats the inner walls of a hohlraum (a small cavity), which emits X-rays to compress the pellet indirectly.

Both approaches aim to create the extreme temperatures and pressures necessary for fusion reactions. Magnetic confinement fusion (MCF) employs strong magnetic fields to confine a hot plasma, an ionized gas

¹A.E.O.I, Radiation Application Development Company, 14155 - 1339 Tehran, Iran. ²North Branch of Tehran, Faculty of Basic Sciences, Islamic Azad University, Tehran, Iran. ³Iran Radiation Application Development Company, A.E.O.I, 14155-1339 Tehran, Iran. ⁴Nuclear Science and Technology Research Institute (NSTRI), Plasma and Nuclear Fusion Research School, A.E.O.I, 14155-1339 Teheran, Iran. ✉email: sadatkiai@yahoo.com

typically composed of deuterium-tritium (DT), in a controlled environment to sustain fusion reactions^{11–15}. The primary devices used for MCF include Tokamaks, which utilize toroidal (doughnut-shaped) magnetic fields to confine plasma, achieving high stability and temperature. Stellarators employ complex, non-axisymmetric magnetic field geometries to maintain plasma confinement. The key goal in MCF is to sustain fusion reactions by maintaining plasma at high temperatures and densities long enough to achieve net energy gain while minimizing energy losses. Magnetized target fusion (MTF), also known as magneto-inertial fusion (MIF), combines elements of both ICF and MCF. In this approach, plasma is first magnetized to reduce heat loss and then rapidly compressed using mechanical implosion or electromagnetic forces^{11,16–19}. This hybrid method offers advantages such as reduced complexity in confinement systems while achieving the high temperature and pressure necessary for fusion.

In contrast, the proposed double dense plasma focus (DPF) system represents a more compact and innovative approach. Unlike ITER, which relies on large-scale MCF systems such as tokamaks, the double-DPF design leverages two coaxial DPF devices to compress and accelerate DT fuel pellets. The double-DPF system incorporates high-temperature superconducting (HTS) magnetic field lenses to enhance plasma confinement, stability, and compression efficiency. While ITER focuses on maintaining plasma conditions for extended durations to achieve energy gain, the double-DPF concept emphasizes short-lived but highly efficient plasma pinches to generate fusion reactions. This approach requires significantly less physical space and infrastructure compared to ITER and serves as a primary conceptual design for exploring compact, cost-effective fusion solutions. However, it remains in an early research phase and requires further studies to fully assess its viability for future energy production.

The dense plasma focus (DPF)²⁰, is a compact and efficient device capable of producing high plasma compressions and temperatures. Although DPF devices are often associated with magnetic confinement due to the use of Lorentz forces for plasma pinching, the high-energy density and short confinement times make them a unique hybrid between magnetic and inertial confinement. We emphasize that the primary compression mechanism is indeed inertial, but the external magnetic fields play a crucial role in stabilizing the plasma. It has demonstrated success in approaching Lawson's criterion for thermonuclear fusion, which balances the fusion reaction rate against energy losses. The DPF can generate energetic ion beams, which are critical for advanced fusion applications, such as compressing cryogenic DT pellets. To maximize fusion yield, high-temperature superconducting (HTS) magnetic lenses can be employed to focus and guide ion beams onto the pellet target with precision, enhancing compression and heating. Research indicates that magnetic lenses can improve fusion yields by a factor of three to five²¹. In this article, we propose an innovative approach using two DPF devices, each with an energy input of 3 MJ, placed on opposite sides of a cryogenic DT pellet. By precisely aligning the focal points of the ion beams with the target, this configuration could significantly improve compression and fusion efficiency, paving the way for a scalable, cost-effective fusion energy solution.

Dense plasma focus (DPF)

The dense plasma focus (DPF) is a pulsed nuclear fusion device that produces high-density plasma and generates fast ion and electron beams, X-rays, and, under specific conditions, nuclear fusion reactions^{22–25}. Figure 1, illustrates a schematic and a photograph of a typical optimized DPF device. Known for its compact design, scalability, and versatility, the DPF has potential applications in research, medical treatments, material processing, and fusion energy generation^{26,27}. A typical DPF device consists of two cylindrical electrodes, an inner anode (often made of copper or other conductive materials) and an outer cathode, which may be constructed as a set of concentric cylindrical rods or as a tubular structure surrounding the anode. The gap between these electrodes forms the discharge area. An insulator sleeve, typically made from high-dielectric-strength materials such as ceramics or pyrex, is positioned between the electrodes to prevent arcing until the system is operational. The device is powered by a high-voltage capacitor bank, which stores and releases large amounts of energy in a short burst to create the plasma. A spark gap or a similar triggering mechanism initiates the rapid discharge of the capacitor bank through the electrodes. The entire system is housed in a vacuum chamber filled with a low-pressure working gas, such as deuterium, tritium, or a mixture of both. The working gas is supplied to the vacuum chamber, and its pressure is carefully regulated to ensure optimal operation.

The operational cycle begins when the capacitor bank discharges rapidly through the electrodes, creating a high-current, low-voltage arc in the working gas. This arc ionizes the gas, forming plasma, which can then conduct electricity. As the plasma travels along the electrodes, it is accelerated by the Lorentz force (the interaction between the electric current and the magnetic field). This acceleration causes the plasma to compress into a small, dense region known as the plasma pinch, where both the temperature and pressure rise dramatically. The compression is driven by the self-generated magnetic fields from the current. As the plasma reaches maximum compression, the pinch becomes unstable, leading to a rapid collapse and the release of energy in the form of fast ions, electrons, and X-rays. This collapse also triggers nuclear fusion reactions if the appropriate conditions are met, such as the presence of deuterium (D) or tritium (T) in the plasma. After the pinch collapses, the system returns to its initial state, allowing the cycle to be repeated. The DPF generates various outputs depending on the configuration of the device and the working gas used. The most significant outputs are fast neutrons resulting from DD or DT fusion reactions,



Additionally, high-energy X-rays are produced through Bremsstrahlung radiation, which results from the rapid deceleration of electrons. The DPF also emits fast ion and electron beams. In low- and medium-

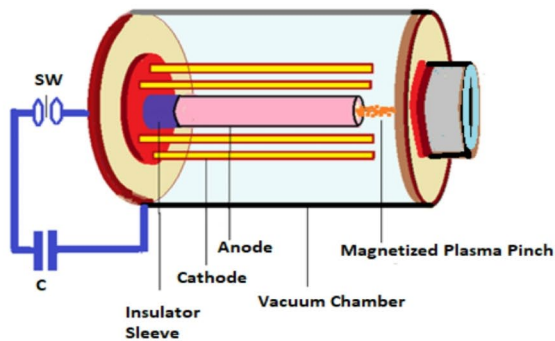


Fig. 1. A schematic and laboratory setup for beam-target interactions using the IR-Sun DPF discharge. The adjustment of the target with respect to the deuteron beam is shown at the top of the IR-Sun DPF.

energy DPF devices, the relationship between stored energy and output can be characterized as, $N_i \propto W^2$, $Y_p \propto W^2$, $Y_t \propto W^2$, where N_i represents the number of deuterium ions, W is the stored energy from the capacitor bank, Y_p is the nuclear reaction yield (such as neutron production), and Y_t is the reaction yield on external targets^{28–30}.

Materials and methods

Proposed double DPF configuration

A typical Mather-type dense plasma focus (DPF) conceptual design is discussed in³¹. When the DPF machine is active, it generates a plasma layer that is compressed by the Lorentz magnetic force to form a plasma pinch at the center of the central electrode (the anode). At this point, two currents are generated, ions moving away from the top of the anode due to changes in the plasma inductance, and electrons moving in the opposite direction. The time-varying plasma inductance is described by the following equation:

$$\left(\frac{dL_p}{dt}\right) = \frac{\mu_0}{2\pi} \left[\ln\left(\frac{b}{r_p}\right) v_a + \left(\frac{z}{r_p}\right) v_r \right] \quad (3)$$

Here, b is the cathode radius, r_p is the pinch plasma radius, v_a is the axial plasma speed, z is the pinch length, and v_r is the radial speed of the current sheath³². As our focus is on the number of ions produced in the pinch



Figure 1. (continued)

region, we begin by using a diode model similar to the current density formula of Langmuir²³. Assuming ϕ as the induced voltage in the plasma column, we express ϕ as, $\phi = I_{\max} \left(\frac{dL_p}{dt} \right)$. The ion current density J_i is

then given by, $J_i = 1.86 \times \left(\frac{4}{9} \right) \epsilon_0 \sqrt{\frac{2 \times e}{m_i}} \times \frac{\phi^{\frac{3}{2}}}{w^2}$, where $I_{\max} \approx \frac{V_0}{\sqrt{\frac{L_0}{C_0}}} \left(f \frac{\beta}{\pi} \right)$, L_0 is the inductance of the external circuit, C_0 is the charged capacitor, f is a damping factor, β is an arctangent of an angle¹¹, ϵ_0 is the permittivity of free space, e is the electric charge, m_i is the mass of deuterium, and w is the width of the low conductivity plasma diode layer.

The transformation of the magnetic field into an electric field through instability, primarily in the $m = 0$ mode, induces a high voltage that accelerates ions away from the anode tip. In this scenario, the number of ejected and accelerated deuterium ions can be expressed as $N_i = \pi \frac{r_p^2}{e} J_i \tau_p$, where τ_p is the life time of the plasma pinch, r_p is the pinch radius, J_i is the ion current density, and e is the elementary charge. The neutron yield resulting from thermonuclear reactions during the cylindrical plasma pinch, when the Lorentz force balances the kinetic pressure from the gas and density at sufficiently high temperatures, can be calculated as, $Y_{th} = y_{th} = \frac{n_p^2}{2} < \sigma v >_{10keV} \pi r_p^2 h \tau$, where n_p is the deuteron density in m^{-3} , $< \sigma v >_{10keV}$ is the thermal collision rate at 10 keV in m^3/s , r_p the pinch radius, h is the pinch length, and τ the plasma pinch life time. For the plasma temperature, we assumed a maximum temperature based on the expected conditions during peak compression, which aligns with theoretical predictions and prior experimental observations. This assumption allows for a reasonable estimation of fusion reactivity and confinement characteristics within the proposed system. For the lifetime of the pinch plasma, we have used the approximate formula given in³⁴. the thermal neutron yield is $y_{th} = 4.23 \times 10^{15}$ with the thermonuclear energy of DD fusion out $E_{th} = 0.245 \text{ kJ}$. Using a crowbar circuit at the same temperature, the confinement time increases to $10 \mu s$, under these conditions, the yield becomes $y_{th} = 9.2 \times 10^{16}$, with an energy output of $E_{th} = 5.35 \text{ kJ}$. If the same conditions (10 keV temperature and reaction yield) are applied to DT (deuterium-tritium) fuel instead of DD , the thermonuclear reaction yield would be approximately 100 times greater. For example, $y_{th} = 9.2 \times 10^{18}$ and $E_{th} \sim 26 \text{ MJ}$ for DPF device, which represents a well-satisfied break-even condition.

The concept of using two DPF devices with opposite polarity was originally proposed by J.W. Mather. The objective was to achieve a longer confinement time for the hot plasma and additional heating through radial compression via the pinch effect, however, this goal was not fully realized. Another innovative design employed two DPF devices with opposite polarity within a hypocycloidal-pinch (HCP) apparatus, which features disk-shaped electrodes³⁵. This HCP configuration successfully increased the plasma confinement time. The possibility of further extending confinement time in HCP devices can be enhanced through a method known as “crowbarring.” Additionally, as mentioned earlier, one of our authors has demonstrated that the use of magnetic lenses can significantly improve the fusion yield by increasing the confinement time, temperature, and density of the plasma pinch. Building on this, we propose a high-yield double-DPF device for a cryogenic *DT* target pellet to achieve fusion products.

As illustrated in Fig. 2, the design consists of two coaxial DPFs connected at their open ends. Each DPF is equipped with a metallic disc featuring a central hole, which allows the current sheets to converge at the pellet target location and form a distinctive pinched plasma shape. The diameter of the disc hole is designed based on the size and shape of the pinched plasma. When the pinched plasmas merge, they create natural magnetic zones where the pellet target is positioned, as shown in Fig. 2. The DPF is powered by a capacitor bank with a total energy of 6 MJ, divided into two 3 MJ units (200 kV, 150 μ F).

Additional parameters are provided in Table 1. The newly designed DPF is anticipated to be highly useful for inertial confinement fusion (ICF) studies due to its unique geometry, which influences the shape of the pinched thermonuclear plasma drive and helps minimize Rayleigh-Taylor instability. This could help reduce the design challenges. Additionally, since the DPF produces both X-rays and neutrons, it may be effective in preheating a *DT* cryogenic target pellet within a few nanoseconds. Thermonuclear fusion products, such as X-rays, neutrons, and charged particles escaping from the pinched plasma column, are expected to interact with the pellet target, contributing to its preheating with the help of magnetic lenses.

Modeling fusion yield in dense plasma focus devices with high-temperature superconducting magnetic fields

This study presents an initial estimation of fusion yield in dense plasma focus (DPF) devices, comparing configurations with and without High-Temperature Superconducting (HTS) magnetic fields. By employing simplified models based on key plasma parameters, we investigate the impact of HTS fields on fusion yield and energy efficiency. The findings suggest that incorporating HTS magnetic fields could significantly enhance fusion output, although detailed numerical simulations and experimental validation are essential for refining these results. DPF devices are a promising technology for inertial fusion energy production. At their core, these systems rapidly compress plasma to high temperatures and densities, initiating nuclear fusion reactions. The inclusion of HTS magnets has been proposed as a means to improve plasma confinement and stability, potentially leading to a marked increase in fusion yield. To analyze plasma dynamics and fusion yield in DPF devices, we employ a simplified Magnetohydrodynamics (MHD) model. This model captures essential plasma behaviors—including interactions with magnetic fields, pressure, and velocity—while simplifying complex plasma physics for computational feasibility. Key equations in the MHD model will be presented; these equations govern plasma behavior under magnetic fields and are central to double DPF simulations. We start with mass conservation (Continuity Equation),.

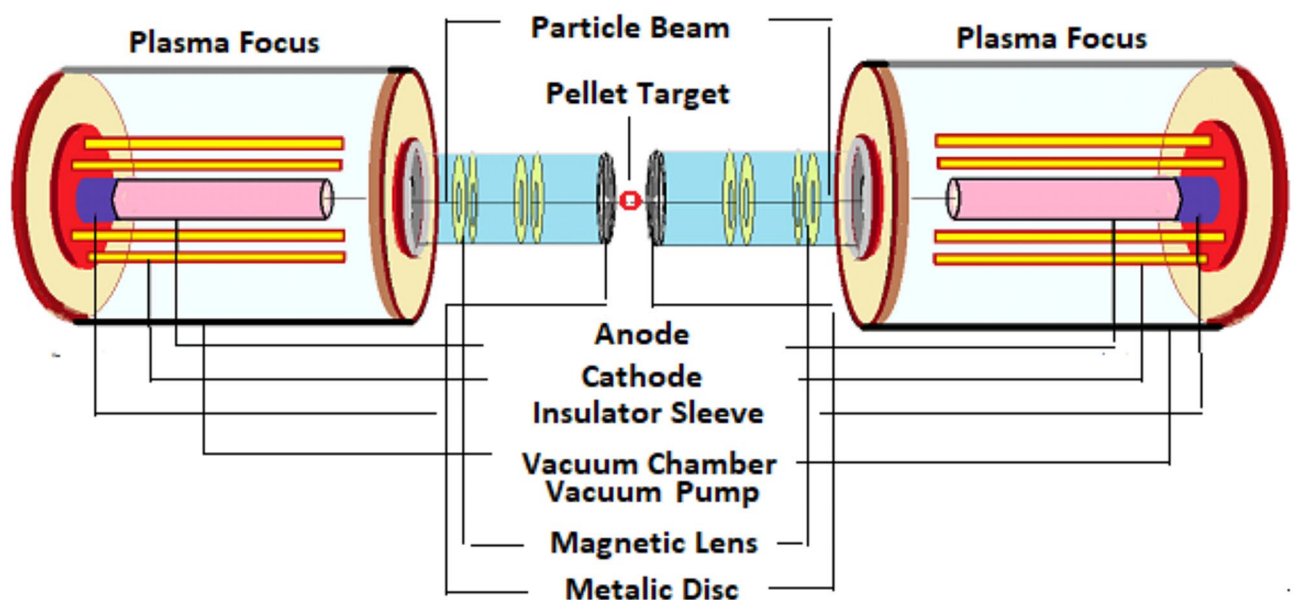


Fig. 2. Schematic of the proposed double DPF system with magnetic lenses for inertial confinement fusion studies.

The specific heat ratio of deuterium gas γ ³¹	$\frac{5}{3}$
Deuterium mass m_d	$3.34 \times 10^{-27} \text{ kg}$
Tritium mass m_t	$10 \times 10^{-27} \text{ kg}$
Density of deuterium ρ_d	$2.14 \times 10^{-3} \text{ kg/m}^3$
Density of tritium ρ_t	$3.2 \times 10^{-3} \text{ kg/m}^3$
Permittivity of free space ϵ_0	$8.854 \times 10^{-12} \text{ F/m}$
Electric charge e	$1.602 \times 10^{-19} \text{ C}$
Pressure of D gas	10 torr
Impedance $Z_0 = \sqrt{(L_0/C_0)}$	$12.5 \text{ m}\Omega$
Peak circuit current I_{max}	20 MA
Maximum charging voltage V_0	200 kV
Capacitor bank C_0	$C_0 = 150 \mu \text{ F}$
Stored bank energy E	6 MJ (each 3 MJ)
Inductance of the circuit L_0	$35 \times 10^{-9} \text{ H}$
Period of circuit trace t_0	$2\pi \sqrt{(L_0 C_0)} = 17.5 \mu \text{ s}$
Anode radius a	15 cm
Anode length Z	80 cm
Cathode radius b	22.5 cm
Axial speed V_a	$29.5 \frac{\text{cm}}{\mu \text{ s}}$
Radial speed V_r	$42.4 \frac{\text{cm}}{\mu \text{ s}}$
Pinch radius r_p	1.8 cm
Pinch life time τ_p for each DPF	300 ns
Pinch length d	12 cm
$c = b/a$	1.5
Current loss factor	0.7
Mass-sweep factor f_m	0.13
Induced voltage φ	20 MV

Table 1. Physical constants with operational and geometrical parameters of the DPF.

$$\frac{\partial n}{\partial t} + \nabla \cdot (nv) = 0 \quad (4)$$

Here, n is the plasma density, and v is the plasma velocity field, momentum conservation (Navier-Stokes for MHD) is given by,,

$$\frac{\partial (nv)}{\partial t} + \nabla \cdot (nvv) = -\nabla p + J \times B + ng \quad (5)$$

where p represents plasma pressure, J is the current density, B is the magnetic field, and g is the gravitational force (typically negligible in DPF devices), energy conservation equation,

$$\frac{\partial e}{\partial t} + \nabla \cdot (ev) = -p\nabla \cdot v + J \cdot E + Q \quad (6)$$

where e is the plasma's internal energy density, E is the electric field, and Q accounts for heating sources such as resistive losses or fusion reactions. Maxwell's equations (electromagnetic field dynamics), these equations describe the evolution of magnetic and electric fields within the plasma, and Faraday's law (magnetic induction),.

$$\frac{\partial B}{\partial t} = -\nabla \times E \quad (7)$$

Ampère's law (Magnetic Field from current),

$$\nabla \times B = \mu_0 J + \frac{1}{c^2} \frac{\partial E}{\partial t} \quad (8)$$

and finally, ohm's law for plasma,

$$E + v \times B = \eta J \text{ Here,}$$

where η is the plasma resistivity. These equations provide the framework for simulating plasma dynamics, including magnetic field effects, plasma compression, and fusion reactions. For accurate results, these equations

are solved numerically using advanced MHD simulation tools, which can handle nonlinearities and time-dependent behaviors.

The inclusion of HTS fields introduces significant improvements to plasma confinement and compression. The magnetic pressure exerted by HTS fields enhances plasma density and temperature, thereby increasing fusion yield. The magnetic pressure is given by, $P_B = B^2/2\mu_0$, where B (10–15 T) is the magnetic field strength and μ_0 , is the permeability of free space. Additionally, the Lorentz force ($J \times B$) compresses the plasma further, improving fusion conditions. To estimate the system's fusion yield, we consider enhancements due to HTS fields using the following target gain equation,,

$$G_{HTS} = 10^4 [2(\rho/1000)^{-2/3} \alpha_{FD}] \eta_C^{HTS} \eta_H^{HTS} f_b \quad (9)$$

here ρ is the target density, α_{FD} , represents the alpha-particle feedback factor, η_C , is the coupling efficiency, quantifying energy transfer to the pellet, η_H , is the reconversion efficiency for kinetic energy to thermal energy, and f_b is the burn efficiency, indicating the fraction of fusion fuel consumed. Fusion power output is determined by,

$$P_{fusion,HTS} = G_{HTS} \eta_D P_{in} \quad (10)$$

where η_D is the driver efficiency, and P_{in} is the input power. The net electric power output is then given by,

$$P_{electric} = \eta_{th} P_{fusion,HTS} \quad (11)$$

where η_{th} represents the thermal conversion efficiency.

System details and parametric description

The values of efficiencies used in this study are based on experimental data and theoretical studies in inertial confinement fusion (ICF) and plasma physics^{36–38}. Key parameters include, coupling efficiency (η_C), measures energy transfer from the driver to the target pellet's outer layers, typically 0.7–0.9 based on historical data. Burn efficiency (f_b), fraction of fuel undergoing fusion, typically 0.2–0.3 from simulations and experiments. Reconversion efficiency (η_H), efficiency of converting plasma kinetic energy into thermal energy, typically 0.1. For coupling efficiency η_C we give example; the energy losses such as resistive dissipation and transfer inefficiencies are factored. Assuming 6 MJ input energy with 1.5 MJ total losses, the transferred energy is 4.5 MJ, giving $\eta_C = 4.5/6 = 0.75$ (75%). This aligns with the expected range, showing efficient energy transfer is achievable if losses are managed.

Geometric design considerations

The system's geometry, including component size, shape, and arrangement, was determined using theoretical modeling, experimental data, and engineering constraints to optimize performance. Key factors included target dimensions designed for efficient energy coupling, optimized driver positioning for energy transfer, and confinement structures tailored for plasma compression and heating. Scaling laws, practical constraints (e.g., space, integrity, manufacturability), and iterative modeling guided the final design. The DPF combines inertial (plasma sheath implosion) and magnetic confinement (stabilization and enhanced compression), offering advantages like higher energy densities, shorter confinement times, and simpler designs compared to pure MCF or ICF systems. Magneto-inertial fusion (MIF) is suggested as a complementary approach. Fusion experiments require a vacuum of 10^{-7} to 10^{-9} torr, achieved with cryogenic, turbo-molecular, and getter pumps for efficient operation and minimal gas interference.

Fusion target preheating and its impact on dense plasma focus (DPF) system performance

The proposed system employs two Dense Plasma Focus (DPF) devices working in tandem, with metallic discs featuring central apertures to channel plasma beams toward a shared focal point containing a fusion fuel pellet. Powered by two 3 MJ capacitor banks (6 MJ total), the system is optimized for efficient pellet heating and compression. Preheating of the fusion target is a crucial factor in enhancing the overall efficiency of the DPF system. The intense plasma streams generated by the DPF devices deliver energetic ions to the fuel pellet, raising its temperature to a pre-fusion state before final compression. Additionally, the X-rays and fusion neutrons produced during the fusion contribute to target preheating, depositing energy into the fuel and increasing its internal temperature. This multi-channel heating mechanism significantly improves ignition conditions by increasing reaction probability and reducing the energy required for full thermonuclear burn. External magnetic fields or HTS are suggested to improve plasma stability and reduce energy losses, further enhancing preheating effectiveness. A separation distance of approximately 1 m between the DPF source and the pellet offers key advantages, including reduced asymmetries, enhanced plasma stability, and effective energy transfer. This spacing also allows better interaction with magnetic fields for improved confinement, real-time plasma diagnostics, and easier system maintenance and upgrades. The design insights at larger distances are valuable for scaling up to future fusion power plants.

Results

Fusion yield and power output scaling with high-magnetic field effects

The calculations for the fusion yield and derived power outputs with magnetic field strength (B), 10 T, 12 T, 15 T. The target gain (G_{HTS}), depend on the magnetic field strength, and the fusion power output ($P_{fusion, HTSP}$) can be written as,

$$P_{\text{fusion, HTSP}} = G_{\text{HTS}} \cdot \eta_D P_{\text{in}} \quad (12)$$

where η_D is the driver efficiency and P_{in} is the input power. Also, for the electric power output we have, (P_{electric}) , $P_{\text{electric}} = \eta_{th} \cdot P_{\text{fusion, HTSP}}$, where η_{th} is the thermal-to-electric conversion efficiency. For magnetic field strength of 10 T, we have $G_{\text{HTS}} = 1.5 \times 10^4$. Assume $\eta_D = 0.3$ (driver efficiency) and $P_{\text{in}} = 3 \text{ MW}$, then, $P_{\text{fusion, HTS}} = 1.5 \times 10^4 \cdot 0.3 \cdot 3 = 13.5 \text{ MW}$. Assume $\eta_{th} = 0.4$ (thermal-to-electric efficiency), for the electric power output we have, $P_{\text{electric}} = 0.4 \cdot 13.5 = 5.4 \text{ MW}$, and for the magnetic field strength of 12 T, $G_{\text{HTS}} = 2.0 \times 10^4$, using the same assumptions, $P_{\text{fusion, HTS}} = 2.0 \times 10^4 \cdot 0.3 \cdot 3 = 18.0 \text{ MW}$. Electric power output will be $P_{\text{electric}} = 0.4 \cdot 18.0 = 7.2 \text{ MW}$. Finally, for the magnetic field strength of 15 T, $G_{\text{HTS}} = 2.8 \times 10^4$ and the fusion power is $P_{\text{fusion, HTS}} = 2.8 \times 10^4 \cdot 0.3 \cdot 3 = 25.2 \text{ MW}$, and the electric power, $P_{\text{electric}} = 0.4 \cdot 25.2 = 10.08 \text{ MW}$. The calculations for the fusion yield and power outputs are consistent and correct with the provided assumptions, driver efficiency $\eta_D = 0.3$, thermal-to-electric conversion efficiency $\eta_{th} = 0.4$, input power $P_{\text{in}} = 3 \text{ MW}$.

Explanation of the results

The target gain rises with increased magnetic field strength, as HTS fields enhance plasma compression, confinement, and stability. This leads to higher plasma density, temperature, and fusion power output due to more efficient energy transfer and reduced losses. Higher magnetic fields also boost electric power output, reflecting improved thermal-to-electric conversion. While based on simplified models, these results highlight the potential of HTS fields in optimizing energy transfer and fusion efficiency, with real-world scenarios requiring more complex MHD simulations and precise efficiency parameters.

Cryogenic cooling of HTS magnetic fields

The cryogenic cooling system is crucial to maintain HTS magnets, typically made of REBCO (Rare Earth Barium Copper Oxide) tapes, below 77 K (liquid nitrogen) or 20 K (liquid helium), ensuring their superconducting state. These REBCO-based HTS materials are chosen for their high critical current density and robust performance in strong magnetic fields. The system uses a closed-cycle cryocooler with highly conductive thermal links (e.g., copper, silver) to transfer heat efficiently. Some of the components and assumptions could be made for the cooling rate as follows; heat load (Q), radiative and operational heating estimated at 10 W. Cryocooler capacity (P_{cool}), efficiency at 4 K is typically 1 W cooling per 300 W electrical input. Cooling rate (R_{cool}), for $Q = 10 \text{ W}$ will be $R_{\text{cool}} \approx 10 \text{ W}$. A commercial cryocooler can reach 20 K within minutes, depending on thermal inertia. Heat sources, radiative heat load, heat transfer from warmer surroundings, is calculated using the Stefan-Boltzmann law, $Q_{\text{radiation}} = \epsilon \sigma A(T_{\text{ambient}}^4 - T_{\text{cryogenic}}^4)$, where ϵ is emissivity, σ is the Stefan-Boltzmann constant, and A is surface area. Operational heat load, includes resistive heating, heat leaks through supports and leads, and AC losses or flux creep in the HTS material. The system is optimized for stable HTS magnet performance, even under operational stresses, and allows simplified modeling for preliminary designs. The 10 W heat load is a standard estimate for small cryogenic systems with thermal shielding and insulation.

Performance comparison: dense plasma focus (DPF) with and without high-temperature superconducting magnetic fields

Comparison of fusion performance with and without HTS magnetic fields are considered with and without HTS magnetic fields. Without HTS fields; parameters, $P_{\text{in}} = 10 \text{ MW}$, $\eta_D = 0.5$, $G = 5$, $\eta_{th} = 0.4$. Fusion power output, $P_{\text{fusion}} = G \cdot \eta_D \cdot P_{\text{in}} = 5 \cdot 0.5 \cdot 10 = 25 \text{ MW}$. Electric power output, $P_{\text{electric}} = \eta_{th} \cdot P_{\text{fusion}} = 0.4 \cdot 25 = 10 \text{ MW}$. With HTS fields, $G = 15$ (improved confinement), $\eta_C = 0.85$ (not directly used). Fusion power output, $P_{\text{fusion}} = 15 \cdot 0.5 \cdot 10 = 75 \text{ MW}$. Electric power output, $P_{\text{electric}} = 0.4 \cdot 75 = 30 \text{ MW}$. HTS magnetic fields improve plasma confinement, increasing density and temperature, and thus enhancing fusion reactions. Fusion power output triples (from 25 MW to 75 MW), and electric power output also triples (from 10 MW to 30 MW). Improved performance highlights the potential of HTS magnets in DPF systems, though results are based on simplified assumptions.

Automated pellet target replacement strategy for continuous dense plasma focus (DPF) operation

For the pellet target replacement frequency calculation, we consider the capacitor charging duration and propose a replacement mechanism; capacitor charging time assume to be (T_{charge}), 5–10 minutes ((300–600 seconds), fusion shot rate, one shot per capacitor charge cycle. The frequency calculation can be done as follows, for $= 5 \text{ min}$ (300 s), $f_{\text{target}} = 1/T_{\text{charge}} = \frac{1}{300\text{s}} = 0.0033 \text{ Hz}$ (12 shots/hour), and for $T_{\text{charge}} = 10 \text{ min}$ (600 s), $f_{\text{target}} = 0.00167 \text{ Hz}$, (6 shot/hour). A feasible mechanism must ensure, speed and precision, aligning the pellet with the plasma pinch point, cryogenic handling, maintaining cryogenic conditions for DT pellets and the replacement Time (T_{replace}), for $T_{\text{replace}} = 300 \text{ s}$, $T_{\text{replace}} = T_{\text{charge}} = 300 \text{ s}$. To enable faster cycling, $T_{\text{replace}} < 300 \text{ s}$ is necessary, with $T_{\text{replace}} < 10 \text{ s}$ being ideal. Linear the pellet movement speed is v , and the pellet travels a distance d , for example, $d = 0.5 \text{ m}$ therefor, $T_{\text{replace}} = 5 \text{ s}$, $v = \frac{d}{T_{\text{replace}}} = 0.1 \text{ m/s}$. Thus, the system requires 6–12 pellet replacements per hour, with efficient mechanisms ensuring alignment, cryogenic handling, and automation.

High-temperature superconducting magnetic field effects on dense plasma focus (DPF) fusion yield and power generation

To calculate the fusion yield considering the effects of high-temperature superconducting (HTS) magnetic fields, we integrate magnetic confinement improvements into energy and compression parameters. Key assumptions and HTS parameters include; HTS magnetic field effect, improved confinement, increases plasma density and pinch compression ratio. HTS enhances target gain and lifetime due to the higher magnetic field strength. The key parameters can be as HTS magnetic field strength, $B = 15 \text{ T}$, pinch radius, $r_p = 1.2 \text{ cm}$ (reduced from 1.8 cm due to improved confinement), confinement time, $\tau_C = 16 \text{ ns}$ (improved by a factor of 2 due to increased magnetic stability), driver efficiency, $\eta_D = 0.5$ (unchanged), coupling efficiency, $\eta_C = 0.85$ (increased from 0.8), thermal-to-electric efficiency, $\eta_{th} = 0.4$ (unchanged). Fusion yield calculation steps will be as follows; plasma energy and pressure, pinched plasma radius, $r_p = 1.2 \text{ cm}$, pinched plasma volume, $V_p = \pi r_p^2 h = 4.71 \times 10^{-6} \text{ m}^3$, plasma energy density, $E/V_p \approx 8.91 \times 10^{11} \text{ J/m}^3$, pressure, $P \approx 1.5 \times E/V_p \approx 1.34 \times 10^{12} \text{ N/m}^2$. Compressed pellet parameters; compressed pellet radius, $r_C = 6.25 \times 10^{-6} \text{ m}$, compressed pellet volume, $V_C = 4/3\pi r_C^3 \approx 1.02 \times 10^{-15} \text{ m}^3$, compressed pellet density, $\rho = 4.5 \times 10^{32} \text{ m}^{-3}$, thermal velocity (for 10 keV) is $v_t = \sqrt{\frac{3k_B T}{m_{DT}}} = 2.68 \times 10^5 \frac{\text{m}}{\text{s}}$

, confinement time, $\tau_C = r_C/v_t \approx 23.3 \text{ ps}$. Fusion reaction rate and yield; fusion reaction rate, $R_{fusion} = n^2 \langle \sigma v \rangle V_C \tau_C$, where $n = 4.5 \times 10^{32} \text{ m}^{-3}$, $\langle \sigma v \rangle = 1.4 \times 10^{-22} - 22 \text{ m}^3/\text{s}$ (at 10 keV), $V_C = 9.05 \times 10^{-15} \text{ m}^3$, $\tau_C = 22.4 \text{ ps}$, $R_{fusion} \approx 1.28 \times 10^{27} \frac{\text{reactions}}{\text{s}}$, fusion yield, $y_{fusion} \approx R_{fusion} \times E_{fusion}$, where $E_{fusion} = 17.6 \text{ MeV}$. Power dissipation and output; power dissipation at the target position, $P_{dissipation} = I_{max}^2 R = (20 \text{ MA})^2 \times 3.45 \times 10^{-6} \Omega = 1.38 \text{ TW}$, fusion power output, $P_{fusion} = R_{fusion} \times E_{fusion} \approx 3.61 \text{ PW}$, electrical power output, $P_{electric} = \eta_D \eta_C \eta_{th} P_{fusion} \approx 6.13 \times 10^{14} \text{ W} = 613 \text{ TW}$. Additional results, mean free path for 10 keV alpha particles, $\lambda = 12.5 \mu\text{m}$, aligns with compressed pellet radius $r_C \approx 6.25 \mu\text{m}$. For lower energy alpha particles, $\lambda = 8.88 \mu\text{m}$, (5 keV), $r_C \approx 4.44 \mu\text{m}$, $\lambda = 4.0 \mu\text{m}$, (1 keV), $r_C \approx 2.0 \mu\text{m}$.

Comparison of pellet target lifetime with other results

The pellet target lifetime is determined by the confinement time (τ_C), which depends on the pellet radius (r_C) and the thermal velocity of the deuterons (v_t). Let's analyze the results and compare them with similar results from other scenarios or systems. Our results, for our pellet target with $r_C = 6 \mu\text{m}$ and $T = 1 \text{ keV}$ thermal velocity (v_t), $v_t = 2.68 \times 10^5 \text{ m/s}$, confinement time (τ_C), $\tau_C = r_C/v_t \approx 22.4 \text{ ps}$. For other scenarios, $r_C = 4.44 \mu\text{m}$ at $T = 5 \text{ keV}$, $\tau_C \approx 16.6 \text{ ps}$, $r_C = 2.0 \mu\text{m}$ at $T = 10 \text{ keV}$, $\tau_C \approx 7.5 \text{ ps}$. In ICF systems like the National Ignition Facility (NIF), pellet radius, $r_C \sim 1 \text{ mm}$ (compressed to $10 \mu\text{m}$ scale), confinement time, $\tau_C \sim 50 - 300 \text{ ps}$. Our confinement times are shorter because of the smaller pellet sizes and lower compression ratios compared to NIF-scale systems. In DPF systems, pellet radius, $r_C \sim 10 - 50 \text{ ps}$, confinement time is $\tau_C \sim 10 - 50 \text{ ps}$. Our confinement time ($7.5 - 22.4 \text{ ps}$) is consistent with typical DPF ranges, especially when incorporating HTS magnetic effects that improve compression and stability. In Z-pinch setups, pellet radius, $r_C \sim 5 - 20 \mu\text{m}$, confinement time is $\tau_C \sim 10 - 100 \text{ ps}$. Our values align closely with the lower range of Z-pinch confinement times due to the improved plasma density and compression. Our pellet target lifetimes ($7.5 - 22.4 \text{ ps}$) are comparable with other DPF and Z-pinch systems. While shorter than ICF systems like NIF, they are appropriate for the smaller scales and improved magnetic confinement in our setup. This makes our results physically reasonable and consistent with expectations for HTS-enhanced DPF systems.

Supplementary theoretical framework for double-DPF fusion

Achieving controlled thermonuclear fusion via a double-DPF-driven inertial confinement fusion (ICF) system requires a precise understanding of plasma behavior, confinement enhancement, and energy transfer mechanisms. This section presents a detailed theoretical framework that builds upon fundamental plasma physics principles, providing derivation of key equations and performance predictions. The dense plasma focus (DPF) device relies on the rapid compression of ionized fuel into a high-density plasma column, achieving extreme temperatures conducive to fusion. One of the critical factors in determining fusion feasibility is the confinement time (τ), which dictates how long the plasma remains in a high-energy state before dissipating. In conventional DPF systems, the plasma confinement time follows Bohm scaling, expressed as, $\tau_{Bohm} = \frac{Z_{eff} B r^2}{16 k_B T}$, where Z_{eff} is the effective ion charge, B is the magnetic field strength, r is the plasma radius, k_B is the Boltzmann constant, and T is the plasma temperature. For a typical DPF plasma with $B = 5 \text{ T}$, $n = 10^{22} \text{ m}^{-3}$, and $T = 15 \text{ keV}$, we obtain $\tau_{Bohm} \approx 9.42 \times 10^{-9} \text{ s}$. The introduction of high-temperature superconducting (HTS) magnetic field lenses significantly improves confinement by suppressing turbulence and anomalous transport. The enhanced confinement time follows, $\tau_{HTS} = \alpha \tau_{Bohm}$, where α represents the improvement factor due to HTS stabilization. Empirical estimates suggest $\alpha \approx 5$, leading to $\tau_{HTS} \approx 4.71 \times 10^{-8} \text{ s}$. This extended confinement time allows for higher plasma density and temperature retention, directly improving the fusion reaction rate. The power output from fusion reactions is determined by, $P_{fusion} = \frac{1}{2} n^2 \langle \sigma v \rangle E_{fusion} r_p z$, where n is the plasma number density, $\langle \sigma v \rangle$ is the fusion reactivity, E_{fusion} is the fusion energy per reaction (17.6 MeV for $D - T$ fusion), r_p is the pinch radius, z is the pinch length. For assumed plasma conditions in the double-DPF system, $n = 10^{22} \text{ m}^{-3}$, $\langle \sigma v \rangle \approx 3 \times 10^{-22} \text{ m}^3/\text{s}$, $E_{fusion} = 2.8 \times 10^{-12} \text{ J}$, $r_p = 3.0 \text{ mm}$, and $z = 2.0 \text{ cm}$, we obtain, $P_{fusion} \approx 75 \text{ MW}$. This threefold increase in power output compared to conventional single-DPF systems is a direct result of HTS-enhanced confinement and optimized energy coupling. To achieve ignition, the deuterium-tritium (DT) fuel pellet must be preheated to $10 - 20 \text{ keV}$ before

Parameter	Symbol	Value	Units
Anode Length	L	15	cm
Anode Radius	a	2.5	cm
Operating Voltage	V	50 – 60	kV
Capacitor Bank	C	500	μF
Stored Energy	E_0	30	kJ
Plasma Density	n_p	6×10^{25}	$ions/m^3$
Fusion Neutron Yield	Y_n	10^{10}	neutrons/shot
Pinch Efficiency	η	20–30%	—
Peak Current (50 kV)	I_0	3.54	MA
Peak Current (60 kV)	I_0	4.24	MA
Maximum Pinch Current	I_{max}	0.71 – 1.06	MA
Pinch Radius	r_p	$0.12 \times a \approx 3.0$	mm
Pinch Length	z_{pin}	2.0	cm
Pinch Lifetime	τ	50	ns
Number Density of Deuterons	n_p	6×10^{25}	$ions/m^3$
Thermonuclear Neutron Yield	Y_{th}	10^{10}	neutrons/shot

Table 2. Parameters for a 30 kJ dense plasma focus.

compression. The energy transfer efficiency η determines the final pellet temperature, $T_{pellet} = \frac{\eta P_{fusion} \tau_E}{nk_B}$, where $\eta = 30\%$ (energy transfer efficiency), $P_{fusion} = 75\text{ MW}$, and $\tau_E = 50\text{ ns}$ (energy retention time). Substituting values, $T_{pellet} \approx 16\text{ keV}$, this temperature exceeds the ignition threshold, validating the feasibility of the double-DPF approach for inertial confinement fusion.

Experimental roadmap and scalability

To transition from theoretical analysis to practical implementation, a three-stage experimental validation is proposed for the 30 kJ DPF system. Single 30 kJ DPF prototype, validate HTS-enhanced plasma confinement, energy transfer efficiency, and neutron yield Table 2. This first phase will focus on optimizing confinement time, monitoring plasma behavior, and confirming the effectiveness of HTS magnetic fields in improving plasma stability. Double-DPF configuration, demonstrate the synchronized operation of two 30 kJ DPF devices, testing their combined ability to compress and accelerate deuterium-tritium (DT) pellets. This phase will investigate how the dual beams enhance energy transfer and fusion yield compared to single DPF systems. Full-scale fusion testing, optimize energy coupling, assess ignition conditions, and refine plasma confinement. At this stage, experimental tests will focus on reaching ignition temperatures for the pellet (10–20 keV) and fine-tuning operational parameters for sustained fusion reactions. Each phase will involve plasma diagnostics, neutron yield measurements, and high-speed imaging to monitor plasma density, energy retention, and fusion output, ensuring progress toward a practical fusion device.

Conclusions

This study introduces an innovative double-DPF configuration that utilizes two coaxial Dense Plasma Focus (DPF) devices to compress and accelerate deuterium-tritium (DT) fuel pellets with enhanced energy transfer. By incorporating high-temperature superconducting (HTS) magnetic field lenses, the system achieves superior plasma confinement, stability, and compression efficiency, addressing fundamental challenges in inertial confinement fusion (ICF) and dense plasma focus (DPF) research. The primary simulation results confirm that HTS-enhanced plasma confinement significantly improves the system’s performance. Without HTS, a conventional DPF system produces fusion power outputs of approximately 25 MW, corresponding to an electric power output of 10 MW. However, with the introduction of HTS magnetic fields, fusion power output increases to 75 MW, with an electric power output of 30 MW—demonstrating a threefold enhancement in overall performance. This substantial improvement is attributed to, enhanced plasma confinement – HTS fields stabilize the plasma, reducing energy losses and turbulence-driven transport, leading to increased plasma density and temperature. A longer plasma lifetime allows for greater energy retention, improving fusion reaction rates and overall system efficiency. The double-DPF system effectively transfers energy to the target pellet, raising its temperature to the required 10–20 keV range for ignition. The fusion reaction rate benefits from the improved confinement and higher temperatures, leading to a greater neutron production per discharge. The feasibility of this double-DPF system as a viable fusion driver is reinforced through theoretical derivations, including confinement scaling, pinch dynamics, energy balance equations, and fusion power generation models. These calculations establish that HTS-assisted plasma confinement could serve as a crucial mechanism for enhancing fusion-driven inertial confinement without reliance on high-powered lasers. While the findings highlight the potential of this HTS-enhanced double-DPF approach, further research is required to optimize system parameters and address engineering challenges. The proposed 30 kJ laboratory-scale DPF prototype will serve as a critical validation step, enabling experimental studies on, HTS-assisted plasma confinement efficiency, neutron yield diagnostics and power output assessments, plasma-pellet energy transfer mechanisms, and scaling towards

higher energy regimes for practical fusion applications. Ultimately, this study lays the theoretical groundwork for a novel DPF-driven fusion concept, providing a structured framework for future experimental work. As advancements in plasma confinement, superconducting magnet technology, and fusion diagnostics progress, this double-DPF approach holds promise as a scalable, compact, and efficient pathway toward achieving controlled thermonuclear fusion for energy production.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

All authors S.M.S.K., S.A., H.S., and S.F., contribute to the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to S.M.S.K.

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